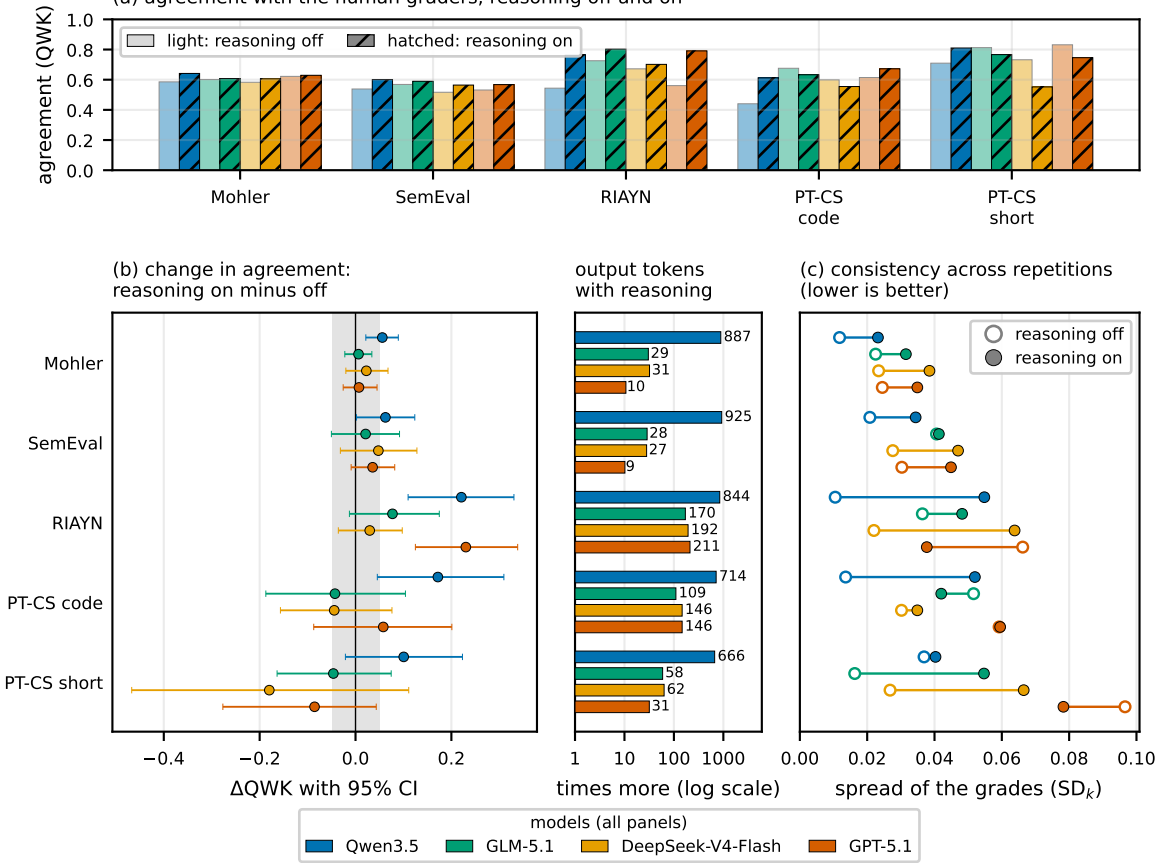
(a) agreement with the human graders, reasoning off and on
agreement (QWK)
1.0
0.8
0.6
0.4
0.2
0.0
light: reasoning off
hatched: reasoning on
Mohler
SemEval
RIAYN
PT-CS code
PT-CS short
(b) change in agreement: reasoning on minus off
Mohler
SemEval
RIAYN
PT-CS code
PT-CS short
−0.4
−0.2
0.0
0.2
ΔQWK with 95% CI
output tokens with reasoning
887
29
31
10
925
28
27
9
844
170
192
211
714
109
146
146
666
58
62
31
1
10
100
1000
times more (log scale)
(c) consistency across repetitions (lower is better)
reasoning off
reasoning on
0.00
0.02
0.04
0.06
0.08
0.10
spread of the grades (SD_k)
models (all panels)
Qwen3.5
GLM-5.1
DeepSeek-V4-Flash
GPT-5.1